THE UNIVERSITY OF BRITISH COLUMBIA
Building Operations

UBC Centre of Comparative Medicine (CCM)
Clayton Boiler Safety Device Check Sheet

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Safety Device Check Sheet Clayton Boiler

This procedure assists in checking the safety devices on the boiler. These safety devices must be tested and calibrated on a Bi-Annual basis

Gas Pressure Switches

The high and low gas pressure switches need to be checked to confirm that they will function properly.

Step	Operation	Cutout Pressure Setting	Cutout Actual Pressure	Date Device Tested
1	Low Gas Pressure Switch	40” WC		
2	High Gas Pressure Switch	100” WC		

Pressure Control Safety Devices

The pressure is controlled in the boiler via the Clayton OUI and is either local or remotely controlled. Under local control the boiler is sensing the pressure control transducer on the separator. When the boiler is under remote control the boiler is sensing the pressure control on the header. The boiler operates with a modulation control and has a 5:1 turndown. The modulating control will stop the firing of the burner and put the boiler through a shutdown cycle. If the modulating control doesn’t work properly there are two high pressure trips and if they don’t work properly there are two safety valves. **Lower the set-points below the operating range to test these devices.**

Operation	Pressure Setting	Pressure at Which it Trips	Date Tested
Test the first High Pressure Trip	130		
Test the second High Pressure Trip	140		

Water Control Safety Devices

This boiler doesn't house a water space and therefor doesn't hold water. The feed water pump is controlled on a VFD which pumps in relation to the boiler firing rate. If the water isn't being delivered at the correct rate there are four different water sensing devices that are temperature controlled. The first two are in the separator and are labeled ATS 1 and ATS 2. The last two are on the coil and are labelled MTCL 1 and MTCL 2. There is one high level switch within the separator which shuts down the boiler in the event that there too much water in the separator. The hot well houses two low level probes. Should the hot well water level go low the low level probes will shut down the boiler.

Devices cannot be tested in place and need to be sent out for calibration

Step	Operation	Temperature Setting	Temperature Cutout - Actual	Date Tested
1	ATS 1 on Separator	380F		
2	ATS 2 on Separator	400F		
3	MTCL 1 on Coil	475F		
4	MTCL 2 on Coil	500F		
5	Separator High Level Switch			
6	Low Hot well Level Turns off Boiler			

Remote & Emergency Shutdown

As part of the ASA this plant can be remotely shutdown. The remote shutdown can be done from the BMS computer as well as the emergency shutdown button on the wall at CCM.

Step	Operation	Initial when Complete	Date and Time when Complete
1	Confirm the remote shutdown can be executed from the BMS computer.		
2	Confirm the boiler can be shutdown via the emergency shutdown button on the wall outside of the boiler room		